

Community detection in geometric graphs: Recent advances until 2026

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Résumé TODO : introduction UGCM, presenter la litterature

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1 Introduction

Community detection consists in finding clusters in a graph where nodes share more connections with nodes of the same cluster than with nodes of other clusters. This broad topic is detailed in [Fortunato, 2010]. In this report, we focus on random geometric graphs (RGG). Adding spatial information allows us to consider local information based on real or abstract proximity. In social networks, two individuals are more likely to be friends if they are geographically close, or if they share similar interests, age, etc. More concrete examples of RGG can be found in wireless networks [Gomez et al., 2015], in biology [Higham et al., 2008], and also in community detection problems [Sankararaman et al., 2020] word embedding, physic,

The main reference model for community detection is the Stochastic Block Model (SBM) [Holland et al., 1983]. Each node carries a hidden community label and edge probabilities depend on whether two nodes belong to the same community or not. This model comes from the Erdős-Rényi graph, where each node connects to others with the same probability [Erdős et al., 1960]. This model is now well understood, see [Abbe, 2018]. Nonetheless, it has some drawbacks, since it is unrealistic to think that every node can connect to every other one with the same probability scale. Around the same time as Erdős-Rényi, the RGG was introduced by [Gilbert, 1961]. Adding geometric information allows us to mimic some emergent behaviours of real networks, such as sparsity [Del Genio et al., 2011], the "small-world" property [Watts and Strogatz, 1998] or transitivity, namely the idea that a friend of a friend is more likely to be a friend.

RGG and its properties are well studied, see [Penrose, 2003] and more recently [Duchemin and De Castro, 2023]. When we add community structure, we are in a different setting. We observe the graph, and sometimes also the node locations, while the community labels remain hidden. The natural question is then whether the latent structure can be detected, and if so, how accurately the labels can be recovered. We are interested in several levels of inference. First,

distinguishability asks whether the observed graph comes from a geometric model with communities or from a null model without community structure. This is a testing problem. It aims to distinguish the presence of a community structure, not to recover the labels. Then, if recovery is possible, one can ask how accurate it can be.

- *Weak/partial recovery* : detecting communities better than random guessing
- *Almost exact recovery* : the fraction of misclassified vertices vanishes as the number of nodes grows.
- *Exact recovery* : all community labels are correctly recovered, up to permutation.

Thus, in this report, community detection is not only understood as producing a convenient partition of the observed graph, but rather as recovering latent community labels under a probabilistic geometric model.

Previously, we emphasized properties that RGG share with real networks due to the geometric information. However, one can also obtain transitivity, and surely more, with attribute-based models [Li et al., 2018], hypergraphs [Ruggeri et al., 2023] and even geometric hypergraphs [Kovács et al., 2025]. We will not consider these models here, in order to keep the focus on the geometric aspect. Moreover, among the problems mentioned above, one is *Distinguishability* which should not be confused with the *Detection Problem* that we do not study here. The latter usually asks for conditions on the parameters defining the RGG under which the model is distinguishable from an Erdős-Rényi graph. Hence, this is not directly a community question. See, [Bubeck et al., 2016] [Liu et al., 2022] [Mao et al., 2026]. Finally, at the beginning of the literature, some works were interested in recovering the latent position of nodes in the geometric space, and considered this as a community detection problem, [Valdivia and De Castro, 2019] [Eldan et al., 2022]. This is a different problem from the one we are interested in, and it's in some sense harder, because one wants to recover the position (a continuous label) of each node rather than a community (a discrete label).
TODO : parler des gaussian-mixture model + trouver ref; voir commentaire latex !

2 Problem setting and inference tasks

Formally, we observe a graph $G := (V, E)$, where V is the vertex set and E is the set of edges between vertices. Throughout the report, n denotes the asymptotic scaling parameter. Depending on the model, the observed number of vertices may be either deterministic or random, but is of order n . We consider a simple undirected graph **Une remarque là dessus? Pourquoi non-orienté est classique comme hypothèse?**, without self-loops. The vertices are partitioned into k communities $C_1, \dots, C_k$, so $V = \bigsqcup_{i=1}^k C_i$. We denote by $\sigma^* : V \rightarrow [k]$ the ground-truth community label function, defined by $\sigma^*(v) = i$ whenever $v \in C_i$. The community structure is hidden, and our goal is to recover it from the observed data.

An estimator is therefore a function of the observations. Depending on the model, the observations may consist only of the graph G , or of the graph together with the node locations $X = (X_v)_{v \in V}$. Accordingly, we write the estimator as $\hat{\sigma} = \hat{\sigma}(G)$ or $\hat{\sigma} = \hat{\sigma}(G, X)$, and we view $\hat{\sigma} : V \rightarrow [k]$ as an estimated labeling. In general, the labels are only identifiable up to a relabeling of the communities. Indeed, permuting the names of the communities does not change the underlying partition of the vertex set. For this reason, the quality of an estimator is measured up to permutation. Denoting by $\mathfrak{S}_k$ the symmetric group on $[k]$, we define

$$A_n(\hat{\sigma}, \sigma^*) := \frac{1}{|V|} \max_{\pi \in \mathfrak{S}_k} \sum_{u \in V} \mathbf{1}\{\hat{\sigma}(u) = \pi(\sigma^*(u))\}$$

Thus, $A_n(\hat{\sigma}, \sigma^*)$ measures the fraction of correctly classified vertices after the best possible relabeling of the communities. We now give more formal definitions of the problems introduced in Section 1.

Distinguishability. Following [Sankararaman and Baccelli, 2018], distinguishability is a hypothesis testing problem. Let $(\mathbb{P}_n)_{n \geq 1}$ be the laws of a geometric random graph model with community structure and let $(\mathbb{Q}_n)_{n \geq 1}$ be the laws of a null geometric model without communities, both defined on the same observation space. Let Ω_n be the observation space. A test is a measurable function $\varphi_n : \Omega_n \rightarrow \{0, 1\}$ where $\varphi_n = 1$ means that we decide in favor of the community model. We say that $(\mathbb{P}_n)$ and $(\mathbb{Q}_n)$ are distinguishable if there exist tests $(\varphi_n)_{n \geq 1}$ and a constant $\gamma > 0$ such that, under the uniform prior on $\{\mathbb{P}_n, \mathbb{Q}_n\}$,

$$\mathbb{P}(\varphi_n \text{ makes the correct decision}) = \frac{1}{2} \mathbb{P}_n(\varphi_n = 1) + \frac{1}{2} \mathbb{Q}_n(\varphi_n = 0) \geq \frac{1}{2} + \gamma - o_n(1)$$

Weak/Partial recovery. We say that weak/partial recovery is achievable if the estimator performs better than random guessing, that's to say in the symmetric case, if there exists a sequence of estimators $(\hat{\sigma}_n)_{n \geq 1}$ and a constant $\alpha > 1/k$ such that,

$$\mathbb{P}(A_n(\hat{\sigma}_n, \sigma^*) \geq \alpha) \xrightarrow{n \rightarrow \infty} 1$$

Almost exact recovery. We say that almost exact recovery is achievable if there exists a sequence of estimators $(\hat{\sigma}_n)_{n \geq 1}$ such that for every $\varepsilon > 0$,

$$\mathbb{P}(A_n(\hat{\sigma}_n, \sigma^*) \geq 1 - \varepsilon) \xrightarrow{n \rightarrow \infty} 1$$

Exact recovery. We say that exact recovery is achievable if there exists a sequence of estimators $(\hat{\sigma}_n)_{n \geq 1}$ such that,

$$\mathbb{P}(A_n(\hat{\sigma}_n, \sigma^*) = 1) \xrightarrow{n \rightarrow \infty} 1$$

Most works in the literature focus on the case $k = 2$. Although several ideas extend beyond this setting, this is also the framework that we adopt in the

following, except when specified otherwise. In that case, it is convenient to encode the labels by $\{-1, +1\}$, instead of $[2] = \{1, 2\}$. Then “up to permutation” reduces to a global sign flip.

Each vertex also belongs to a space $\mathcal{X}$. In the following, we specify each time whether this space is latent or observed. For theoretical analysis, $\mathcal{X}$ is often chosen for mathematical convenience. The Euclidean space $\mathbb{R}^{d-1}$ is a natural reference because of its symmetry, homogeneity, isotropy, and direct geometric interpretation. See [Abbe et al., 2021], **Citer ceux qui sont euclidiens. In the following, d is an integer.** We also encounter the torus, mainly to avoid boundary effects and to focus on the core ideas of the proofs, **Todo citer.** Lastly, for latent models, the high-dimensional unit sphere $\mathbb{S}^{d-1}$ is a standard reference space to model similarity, and it also provides convenient spectral properties [Castro et al., 2020]. A summary of theoretical works on $\mathbb{S}^{d-1}$ is given in [Duchemin and De Castro, 2023].

In Section 3 we present models used throughout the literature. Hence, we suggest a model to unify them while remaining close to the notation used in the papers. We call it the *Unified Geometric Community Model* (UGCM) and it is defined as,

$$\text{UGCM}(\mathcal{X}, \rho, W_n, \mu_n, Z, \pi, \mathcal{Y}, K_n)$$

where $\mathcal{X}$ is the ambient space and ρ is its metric. $W_n \subset \mathcal{X}$ is the window where nodes are sampled at the scale n . μ_n governs the spatial sampling mechanism. Usually, the vertices are sampled either as n i.i.d. points in W_n with law μ_n or as a Poisson Point Process (PPP) on W_n with intensity measure μ_n . We denote by Z the label set, and by $\pi = (\pi_i)_{i \in Z}$ the distribution of the labels. For all $u \in V$ and independently, $\sigma^*(u) \sim \pi$. $\mathcal{Y}$ the space of pairwise observation. Let $\mathcal{P}(\mathcal{Y})$ denote the set of probability distributions on $\mathcal{Y}$, we consider the kernel observation,

$$K_n : \mathcal{X} \times \mathcal{X} \times Z \times Z \rightarrow \mathcal{P}(\mathcal{Y})$$

and for each unordered pair of vertices u, v , we observe,

$$Y_{uv} \mid (X, \sigma^*) \sim K_n(X_u, X_v, \sigma^*(u), \sigma^*(v))$$

The kernel is the core of the model. We note that Soft-RGG introduced by [Penrose, 2016] and developed in [Duchemin and De Castro, 2023] can be seen as a special case of UGCM where there exists a function $f_n : \mathbb{R}^+ \rightarrow [0, 1]$ such that for every $x, y \in \mathcal{X}$ and $z, z' \in Z$, $K_n(x, y, z, z') = \text{Ber}(f_n(\rho(x, y)))$. Moreover, a particular case of Soft-RGG are the usual random geometric graph, i.e. the hard-threshold special case, when f_n is an indicator function, i.e. there exists $r_n > 0$ such that $K_n(x, y, z, z') = \delta_{\mathbf{1}_{\{\rho(x, y) \leq r_n\}}}$. Then, this framework also accommodates weighted graphs, since the observation space $\mathcal{Y}$ is not necessarily binary.

As we will see, several phase-transition phenomena depend on the connectivity scale of the graph, which is usually measured through the average degree. Let D_n denote the degree of a vertex chosen uniformly at random. We can distinguish three regimes of connectivity, depending on the growth of $\mathbb{E}(D_n)$ as

$n \rightarrow \infty$. The most realistic but also the most delicate regime is the *sparse regime*. Each vertex only carries a finite amount of local information, the expected degree remains bounded,

$$\mathbb{E}(D_n) = \Theta(1)$$

In the *logarithmic regime*, the graph becomes sufficiently connected for exact recovery to become meaningful,

$$\mathbb{E}(D_n) = \Theta(\log n)$$

Finally, the *dense regime* corresponds to the easiest setting, the graph carries a lot of information,

$$\mathbb{E}(D_n) = \Theta(n)$$

3 Geometric community models and recovery thresholds

3.1 Geometric Block Model (GBM)

Historically, one of the earliest geometric models for community detection is the Geometric Block Model (GBM) introduced in [Galhotra et al., 2018]. They mainly study the case of two symmetric communities. In its general form, the model places each vertex at a latent position on the unit sphere $\mathcal{X} := \mathbb{S}^{d-1}$, sampled uniformly, and edges are drawn through a hard geometric rule. For $d \in \mathbb{N} \setminus \{0, 1\}$, $r_{in}, r_{out} \in [-1, 1]$, $r_{ij} := r_{in}\delta_{ij} + r_{out}(1 - \delta_{ij})$, they define

$$\text{GBM}(r_{in}, r_{out}, d) := \text{UGCM}(\mathcal{X}, \|\cdot\|_2, \mathcal{X}, \text{Unif}(\mathcal{X}), \{-1, +1\}, \text{Ber}(1/2), \{0, 1\}, K_n)$$

with $K_n(X_u, X_v, Z_u, Z_v) = \mathbf{1}\{\langle X_u, X_v \rangle \geq r_{Z_u Z_v}\}$. The condition $r_{in} > r_{out}$ ensures that edges within communities are denser than across communities. The space $\mathcal{X}$ is assumed latent, so recovery is done only from the adjacency matrix.

They study the one dimensional circular case, corresponding to $\mathbb{S}^1 \subset \mathbb{R}^2$, hence $d = 2$ in the notation above, in the regime

$$r_{in} := a \frac{\log n}{n} \quad r_{out} := b \frac{\log n}{n} \quad a > b > 0$$

This corresponds to the logarithmic regime. This regime matches the connectivity scale of one dimensional random geometric graphs. Their main algorithm is motif based, that is, for an observed edge $(u, v) \in E$, they count the number of common neighbors of u and v , so the number of triangles containing this edge. Then they use this statistic to decide whether u and v belong to the same community. The geometry creates strong local dependence and transitivity. Unlike in the SBM, triangles are informative in the GBM. Still in this regime, the simple motif counting procedure already yields exact recovery with high probability when $r_{in} \geq 4r_{out}$. The algorithm is efficient, and the journal version later clarifies that this one threshold guarantee from the conference paper is order optimal up to constants but restricted to that parameter range [Galhotra et al., 2018, 2023].

Firstly, [Galhotra et al., 2020] introduces the random annulus graph (RAG). Let $I := \left[b \left(\frac{\log n}{n} \right)^{1/d}, a \left(\frac{\log n}{n} \right)^{1/d} \right]$, they work on $\mathbb{S}^d$ with $RAG_d(n, I)$, a RGG where nodes are connected only if the distance between them falls inside I , hence the name annulus graph. In dimension 1, they obtain a sharp connectivity threshold for RAG : the graph is connected with high probability if $a > 1$ and $a - b > 1/2$, and disconnected with high probability if $a < 1$ or $a - b < 1/2$. For $d > 1$, they give only sufficient conditions [Galhotra et al., 2020].

Secondly, the main algorithmic refinement is to replace the original one threshold triangle rule by a two threshold procedure. The reason is that, once one leaves the regime $r_{in} \geq 4r_{out}$, a single threshold is no longer sufficient. For inter community edges, the normalized triangle count is concentrated around $2r_{out}$, whereas for intra community edges it depends on the latent distance and may lie either above or below that value. They therefore define two thresholds

$$E_S := (2b + f_1) \frac{\log n}{n} \quad E_D := (2b - f_2) \frac{\log n}{n}$$

where f_1 and f_2 are chosen from Chernoff type large deviation inequalities. Edges whose triangle count falls inside the ambiguous interval $[E_D, E_S]$ are removed. This deletes all inter community edges with high probability, while preserving enough informative intra community edges. It remains to show that the surviving intra community edges still connect each community, and this is exactly where the RAG connectivity results are used.

As a consequence, they obtain exact recovery in dimension 1 with explicit conditions and an impossibility result when $a < 1$ or $a - b < 1/2$. However, the exact recovery threshold for the latent one dimensional GBM is still not fully characterized, see Figure 1. A useful contrast is that this gap disappears when the latent positions are revealed. In that case, the one dimensional GBM has a sharp exact recovery threshold which coincides with the sharp connectivity threshold of the RAG. In higher dimension $d > 1$, they also prove a polynomial time exact recovery result at the correct order of magnitude,

$$r_{out} = \Theta \left(\left(\frac{\log n}{n} \right)^{1/d} \right) \quad \text{and} \quad r_{in} - r_{out} = \Theta \left(\left(\frac{\log n}{n} \right)^{1/d} \right)$$

are sufficient, while recovery becomes impossible if one of these quantities is of smaller order [Galhotra et al., 2023].

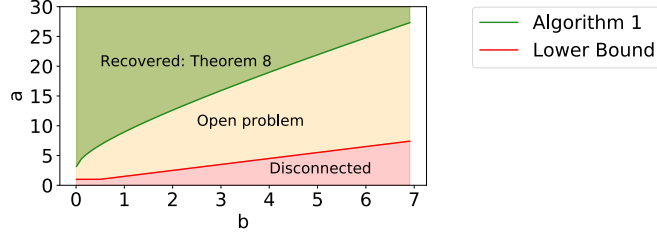


FIGURE 1. Phase diagram for exact recovery in the one dimensional latent GBM in the regime $r_{in} = a \log n/n$ and $r_{out} = b \log n/n$, reproduced from Galhotra et al. [2023], Figure 5. The red region corresponds to disconnected graphs, hence an impossibility result. The yellow region remains an open problem, and the green region corresponds to the parameter range certified by Algorithm 1 (Theorem 8).

Finally, [Chien et al., 2020] studies active learning in the latent GBM with two communities. The model is the same, but the learner is now allowed to query the labels of a small number of carefully chosen vertices to a perfect oracle. Their approach keeps the motif counting idea as a first step, in order to remove many inter community edges, and then uses label queries in a second step. They propose two algorithms. The first combines this pruning step with the graph based active learning procedure S^2 [Dasarathy et al., 2015], while the second is more aggressive and then queries one label per connected component. Their main result is that exact recovery becomes possible with high probability using only a sublinear number of queries, even in parameter regimes where the best known unsupervised guarantees fail. This gives a smooth trade off between the hardness of clustering and the query complexity. They do not prove any optimality result for this active setting. Moreover, the theoretical bounds seem somewhat pessimistic. Although the analysis favors the first algorithm, the paper notes that the second often uses fewer queries in practice, which suggests that the proof can likely be improved [Chien et al., 2020].

3.2 Geometric Stochastic Block Model (GSBM)

In [Sankararaman and Baccelli, 2018] introduces the Geometric Stochastic Block Model (GSBM) and presents in [Abbe et al., 2021] in more detailed way their model. The geometric position are observed, the space $\mathcal{X} := \mathbb{R}^d$ is not latent. Here, n is a scaling parameter and not the number of nodes. Points are sampled as a PPP on a window $W_n := \left[-\frac{n^{1/d}}{2}, \frac{n^{1/d}}{2}\right]^d$ with intensity measure $\mu_n := \lambda \text{Leb}|_{W_n}$ with $\lambda > 0$. The window W_n has a volume n . They study the case of two symmetric communities. Nodes are connected according to their distance and labels through $f_{in}^{(n)}$ and $f_{out}^{(n)}$. We assume for all $r \geq 0$, $1 \geq f_{in}^{(n)}(r) \geq f_{out}^{(n)}(r) \geq 0$ and $f_{in}^{(n)} \neq f_{out}^{(n)}$ upto Lebesgue measure 0. More assumptions on these functions

are given according to the regime that we are interested. The model is then defined as,

$$\text{GSBM}(f_{in}^{(n)}, f_{out}^{(n)}, \lambda, d) := \text{UGCM}(\mathcal{X}, \rho_n, W_n, \mu_n, \{-1, +1\}, \text{Ber}(1/2), \{0, 1\}, K_n)$$

where the metric is the Euclidean distance in the sparse regime, $\rho_n := \|\cdot\|_2$, and toroidal metric in the logarithmic regime, $\rho_n := \|\cdot\|_{T_n}$. The latter is to avoid boundary effects on W_n due to the infinity of neighbors a points can have, for the sparse regime this consideration is negligible. Moreover, the kernel is defined as,

$$K_n : (x, y, z, z') \mapsto \text{Ber} \left(f_{in}^{(n)}(\|x - y\|) \mathbf{1}\{z = z'\} + f_{out}^{(n)}(\|x - y\|) \mathbf{1}\{z \neq z'\} \right)$$

In the sparse regime, nodes have a finite expected degree. Accordingly, we assume that the connection functions do not depend on n , namely, $f_{in}^{(n)} = f_{in}$ and $f_{out}^{(n)} = f_{out}$, and that

$$\int_{\mathbb{R}^d} f_{in}(\|x\|_2) + f_{out}(\|x\|_2) dx < \infty$$

The metric is Euclidean, which allows the finite graphs to be viewed as restrictions of a single infinite random graph [Last and Penrose, 2018]. In this regime, the main question is weak recovery. By monotonicity, there exists a critical value $\lambda_c \in [0, \infty]$ separating the unsolvable and solvable phases. The first result is an impossibility criterion based on the auxiliary random connection model $H_{\lambda, g(\cdot), d}$ defined on the same PPP of intensity λ where two points at distance r are connected independently with probability $g(r)$. Denoting by $\theta(H_{\lambda, g(\cdot), d})$ the percolation probability, the probability that the connected component of a typical node is infinite. They prove that if $\theta(H_{\lambda, f_{in} - f_{out}, d}) = 0$ then weak recovery is impossible. In other words, if the informative part $f_{in} - f_{out}$ does not generate an infinite connected component, then the community information cannot propagate at long range. In particular, for $d = 1$, weak recovery is impossible for every $\lambda > 0$. For $d \geq 2$, this yields the explicit lower bound

$$\lambda \leq \left(\int_{\mathbb{R}^d} (f_{in}(\|x\|_2) - f_{out}(\|x\|_2)) dx \right)^{-1} \implies \text{weak recovery is impossible}$$

On the positive side, they introduce the Good Bad Grid (GBG) algorithm. Although the graph is globally sparse, it remains locally dense because geometry creates many short loops and common neighbors. Hence, for two nearby nodes, the number of common neighbors carries information on whether they belong to the same community. The algorithm first compares nearby pairs using such local statistics, then partitions the space into small cells and propagates consistent decisions across the grid. Its analysis relies on a coupling with a dependent site percolation process. They prove that if $d \geq 2$ then there exists $\lambda_{upper} < \infty$ such that weak recovery is solvable for all $\lambda \geq \lambda_{upper}$. Combined with the lower bound above, this implies $0 < \lambda_c < \infty$ so the sparse regime exhibits a non-trivial

phase transition [Abbe et al., 2021]. In some special cases, they obtain a sharp characterization of the threshold. If connection functions are hard thresholds, that is, if there exist $R_{in}, R_{out} \geq 0$ such that,

$$f_{in} : r \mapsto \mathbf{1}_{\{r \leq R_{in}\}} \quad \text{and} \quad f_{out} : r \mapsto \mathbf{1}_{\{r \leq R_{out}\}} \quad \text{and} \quad R_{in} \geq R_{out}$$

then weak recovery is solvable if and only if $\theta(H_{\lambda, f_{in}-f_{out}, d}) > 0$. So, in this hard threshold case, the phase transition is fully characterized. More generally, they also show a weak form of asymptotic optimality for GBG, when the signal is strong enough, the fraction of correctly classified vertices can be made arbitrarily close to 1. In our notation, this means that for every $\varepsilon > 0$, there exists $\lambda_0 > 0$ such that for all $\lambda \geq \lambda_0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(A_n(\hat{\sigma}_n^{GBG, \lambda}, \sigma^*) \geq 1 - \varepsilon) = 1$$

paragraphe : Distinguability paragraphe : logarithmic paragraphe ouverture

3.3 (Geometric Kernel Block Model, GKBM)

[Avrachenkov et al., 2024] [Allem et al., 2025]

3.4 Geometric Hidden Community Model (GHCM)

[Gaudio et al., 2024b] [Gaudio et al., 2024a] [Gaudio and Guan, 2025] [Gaudio and Jin, 2026]

4 Spectral

(A confirmer) [Chen et al., 2016]
 (papier SBM mais théorie intéressante) [Bordenave et al., 2015]
 Cadre spectral intéressant pour certains espaces [Méliot, 2019]
 Méthode spectrale + Robustesse (la perturbation est géométrique) [Péché and Perchet, 2020]
 Méthode spectrale sur Soft Geometric Block Model (SGBM) (Generalisation de Geometric SBM) [Avrachenkov et al., 2021]

5 Ouvertures

- Considérer des modèles hyperboliques, pas seulement labels & distance-dependant, car ceux-ci captent mieux les propriétés habituelles des réseaux
 - Clustering [Krioukov, 2016]
 - Degré inhomogène [Barabási and Albert, 1999]
 - "Communities with specific mixing patterns" (mesoscale) [Toivonen et al., 2006]
- Communautés apparaissent dans un espace hyperbolique latent [Bruno et al., 2019] Réponse 6ans plus tard : pas toujours! + définition d'un modèle pour "régler" ça [Guarino et al., 2025]
- [Li et al., 2018]

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